IPCON® EngFila Support

Support Filament for Engineering Filaments

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	305 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or PC (max heat resistance: 130 °C)
Spool Size	Outer cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm
Recommendation for body protection during use	This material contains glass fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation.

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.36 g/cm³
Shrinkage Rate*	Specially-defined	0.37% - 0.50%
Surface Hardness	Shore hardness	83D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	1.33%
Melt Index	290 °C, 2.16 kg	3.2 g/10 min
Melting Temperature	DSC, 10 °C/min	245 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	Not annealed: 87 °C Annealed: 158 °C*
Heat Deflection Temperature	ISO 75, 0.45 MPa	Not annealed: 150 °C Annealed: 152 °C*
Heat Deflection Temperature	ISO 75, 1.80 MPa	Not annealed: 65 °C Annealed: 70 °C*
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	75 - 90 °C (Related to load, size, shape, wall thickness, rate of water content, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	26 MPa
Tensile Strength Z	ISO 527, GB/T 1040	8 MPa
Young's Modulus XY	ISO 527, GB/T 1040	2574 MPa

Young's Modulus Z	ISO 527, GB/T 1040	1286 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	9.3%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	4.2%
Flexural Strength XY	ISO 178, GB/T 9341	70 MPa
Flexural Strength Z	ISO 178, GB/T 9341	18 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	2891 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	1763 MPa
Impact Strength XY	ISO 179, GB/T 1043	23.5 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	2.3 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 295 °C, bed temperature: 100 °C, printing speed: printing speed: 100 mm/s) and IPCON PAHT GF (red) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The XY specimens' annealing parameter is: 70 °C, 4 h → 90 °C, 8 h. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Wide printable nozzle temperature range (270 - 330 °C), high heat resistance, high strength and stiffness, and excellent structural stability; suitable for most kinds of engineering materials, easy to peel off, and has little residue; compatible with dual-nozzle and single-nozzle + AMS or CFS for printing with different materials; available in black and white.
2. Compatible with most kinds of engineering materials such as ABS, ASA, PET, PC, PA, PPA, PPS and their carbon fiber, glass fiber reinforced or other modified materials.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 120 °C, 8 - 12 h (Blast drying oven)
Printing Temperature and Humidity	≤ 80 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Body Material	Most engineering materials such as ABS, ASA, PET, PC, PA, PPA, PPS and their carbon fiber, glass fiber reinforced or other modified materials
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	270 - 330 °C
Bed Temperature	80 - 120 °C
Fan Speed	0% - 30%
Max Print Speed	250 mm/s

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